THE WELL-NIGH-AFFINE INVARIANT-RING MODULI SPACE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3424-3433 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 3435-3514, the complete author proof (78 body lines).

Editorial note. Navigation stubs: 16 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 12 of queue_stacks-stacks-more-morphisms.json (pb=3365, stmt 3424-3433, proof 3435-3514).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 09EF: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3514–3514. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03BJ: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3514–3514. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03BK: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3514–3514. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03BL: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3514–3514. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0DTG: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3514–3514. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03BE: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3514–3514. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-categorical-quotient-compare: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3247–3252. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-quotient-compare: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3217–3224. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-well-nigh-affine: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3339–3346. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01WM: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3514–3514. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01I1: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3514–3514. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 050M: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3514–3514. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CQK: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3514–3514. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DTZ: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3514–3514. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 050X: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3514–3514. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DTN: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3514–3514. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let the algebraic stack $\mathcal{X}$ be well-nigh affine. There exists a uniform categorical moduli space*

$$f : \mathcal{X} \longrightarrow M$$

in the category of affine schemes. Moreover f is separated, quasi-compact, and a universal homeomorphism.

Proof. Write $\mathcal{X} = [U/R]$ with (U, R, s, t, c) as in Lemma lemma-well-nigh-affine. Let C be the ring of R-invariant functions on U, see Groupoids, Section 03BE. We set $M = \mathrm{Spec}(C)$. The R-invariant morphism $U \to M$ corresponds to a morphism $f : \mathcal{X} \to M$ by Lemma lemma-quotient-compare. The characterization of morphisms into affine schemes given in Schemes, Lemma 01I1 immediately guarantees that $\phi : U \to M$ is a categorical quotient in the category of affine schemes. Hence f is a categorical moduli space in the category of affine schemes (Lemma lemma-categorical-quotient-compare).

Since $\mathcal{X}$ is separated by Lemma lemma-well-nigh-affine we find that f is separated by Morphisms of Stacks, Lemma 050M.

Since $U \to \mathcal{X}$ is surjective and since $U \to M$ is quasi-compact, we see that f is quasi-compact by Morphisms of Stacks, Lemma 050X.

By Groupoids, Lemma 03BJ the composition

$$U \to \mathcal{X} \to M$$

is an integral morphism of affine schemes. In particular, it is universally closed (Morphisms, Lemma 01WM). Since $U \to \mathcal{X}$ is surjective, it follows that $\mathcal{X} \to M$ is universally closed (Morphisms of Stacks, Lemma 0CQK). To conclude that $\mathcal{X} \to M$ is a universal homeomorphism, it is enough to show that it is universally bijective, i.e., surjective and universally injective.

We have $|\mathcal{X}| = |U|/|R|$ by Morphisms of Stacks, Lemma 0DTZ. Thus $|f|$ is surjective and even bijective by Groupoids, Lemma 03BL.

Let $C \to C'$ be a ring map. Let (U', R', s', t', c') be the base change of (U, R, s, t, c) by $M' = \mathrm{Spec}(C') \to M$. Setting $\mathcal{X}' = [U'/R']$, we observe that $M' \times_M \mathcal{X} = \mathcal{X}'$ by Quotients of Groupoids, Lemma 0DTG. Let C^1 be the ring of R'-invariant functions on U'. Set $M^1 = \mathrm{Spec}(C^1)$ and consider the diagram

$$\begin{array}{ccc} \mathcal{X}' & \longrightarrow & \mathcal{X} \\ \downarrow{\scriptstyle f'} & & \\ M^1 & & \downarrow{\scriptstyle f} \\ \downarrow & & \\ M' & \longrightarrow & M \end{array}$$

By Groupoids, Lemma 03BK and Algebra, Lemma 09EF the morphism $M^1 \to M'$ is a homeomorphism. On the other hand, the previous paragraph applied to (U', R', s', t', c') shows that $|f'|$ is bijective. We conclude that f induces a bijection on points after any base change by an affine scheme. Thus f is universally injective by Morphisms of Stacks, Lemma 0DTN.

Finally, we still have to show that f is a uniform moduli space in the category of affine schemes. This follows from the discussion above and the fact that if the ring map $C \to C'$ is flat, then $C' \to C^1$ is an isomorphism by Groupoids, Lemma 03BK. □